

Compton Frequency Cross-Ratios on Bruhat-Tits Trees: A Pre-Registered Search for Adelic Structure in the Standard Model Mass Spectrum (Version 2.0)

Rowan Brad Quni

2026-07-22

Author: Rowan Brad Quni | **Date:** 2026-07-22 | **License:** QNFO-ULA:
<https://legal.qnfo.org/>

Abstract

Version 1.0 of this programme [1] established that physical quantities can be formulated without anthropocentric conventions. Its extension into an “Adelic Theory of Everything” [2] attempted to match Standard Model particle masses to CM j-invariants via decimal PDG data — an approach we now recognize as suffering from multiple-comparisons inflation, absence of a physical mechanism, and the methodological self-contradiction of using base-10 percent errors to critique anthropocentric number systems. Version 2.0 corrects these failures: we define particle masses as Compton angular frequencies $\omega_i = m_i c^2 / \hbar$ (in natural units, $c = \hbar = 1$), form projective-invariant cross-ratios $\text{CR}(a, b, c, d) = (\omega_a - \omega_c)(\omega_b - \omega_d) / (\omega_a - \omega_d)(\omega_b - \omega_c)$, and test for p -adic valuation structure on the Bruhat-Tits tree $\mathcal{T}_p$ as a geometric signal of adelic physics. All tests were pre-registered on 2026-07-22 before any computation. Of five pre-registered cross-ratios tested at three primes ($p = 2, 3, 5$), two were found to approximate simple rationals (976/919 and 430/419) at precision marginally better than the Dirichlet theorem guarantee. No test exceeds the Bonferroni-corrected significance threshold. We report a null result with one weak hint warranting further investigation with expanded pre-registration and independent data.

1. Introduction: Version 1.0 Post-Mortem

1.1 What was claimed

Version 1.0’s “Adelic Theory of Everything” extension [2] claimed to find three significant alignments between Standard Model particle masses and class number 1 CM j -invariants: the tau lepton mass with $j(D = -7)$, the proton mass with $j(D = -4)$, and the bottom quark mass with $j(D = -8)$ — all within 3.5% in decimal ratio.

1.2 What was actually done

These “discoveries” were produced by:

1. **Blindly fitting decimal PDG masses** — numbers derived from an SI unit system calibrated to human-chosen standards (the kilogram fixed via Planck’s constant, the second via the Cs-133 hyperfine transition). These are decimal floating-point artifacts, not physical invariants.
2. **Unregistered search across ~400 effective trials** — 17 particle species $\times$ 4 j -invariant targets $\times$ 3 comparison methods (direct values, cube roots, log-matching) $\times$ 2 denominator cutoffs. With a 3.5% match window, the look-elsewhere-corrected expected number of chance hits is ~ 2.5 . Finding 3 is unremarkable.
3. **No mechanism.** Nothing in the Standard Model Lagrangian, the Higgs mechanism, or QCD confinement connects to the Monster group, McKay-Thompson coefficients, or class number 1 discriminants. Numbers were compared to other numbers.
4. **Methodological self-contradiction.** A paper arguing against anthropocentric bases, number systems, and human-chosen conventions then used base-10 percent errors, base- e natural logarithms, and decimal PDG masses as its analysis tools.

1.3 What version 2.0 changes

v1.0 Failure	v2.0 Fix
Decimal PDG masses (SI artifacts)	Compton angular frequencies $\omega_i = m_i c^2 / \hbar$ in natural units
Single-particle mass ratios (not invariants)	Cross-ratios $\text{CR}(a, b, c, d)$ — genuine projective invariants under $\text{PGL}(2, \mathbb{Q}_p)$
Unregistered, arbitrary search	Pre-registered on 2026-07-22; exactly 5 cross-ratios $\times$ 3 primes = 15 tests
No physical mechanism	Bruhat-Tits tree geometry: $\text{ord}_p(\text{CR})$ = signed graph distance on $\mathcal{T}_p$, providing a geometric interpretation

v1.0 Failure	v2.0 Fix
Base-10 percent errors	p -adic valuation comparison — the natural ultrametric distance
Claimed “discoveries”	Honest null result with explicit statistical thresholds

2. Physics Foundation: The Compton Frequency as the Fundamental Invariant

2.1 Mass as frequency

In natural units ($c = \hbar = 1$), every particle mass IS its Compton angular frequency:

$$\omega_i = m_i c^2 / \hbar = m_i \quad (\text{in natural units})$$

The electron Compton frequency is the natural reference scale:

$$\omega_e = m_e c^2 / \hbar = 7.763 \times 10^{20} \text{ rad} \cdot \text{s}^{-1}$$

The Zitterbewegung — the rapid oscillatory motion predicted by the Dirac equation as interference between positive- and negative-energy components — has frequency $\omega_{\text{ZBW}} = 2\omega_C$. Every fermion’s mass determines its ZBW frequency. Every ratio m_i/m_e is identically the ratio of Compton angular frequencies ω_i/ω_e — a genuinely dimensionless, frame-independent, unit-independent physical invariant.

Critically, this is NOT the same as comparing decimal numeric magnitudes from an SI table. The Penning-trap cyclotron frequency-ratio metrology that determines m_e to $\sim 1.7 \times 10^{-10}$ relative uncertainty measures $\omega_c = qB/m \Rightarrow m_e/m_{\text{ref}} = \omega_c(\text{ref})/\omega_c(e)$ — a ratio of frequencies in the same magnetic field, never an absolute “mass in kilograms.” The frequency ratio IS the physical quantity; the decimal mass value is a derived convenience.

2.2 Why single ratios are not projective invariants

A single ratio $m_i/m_e = \omega_i/\omega_e$ depends on which particle is chosen as reference. The electron is not privileged by nature — it is privileged by our choice. Under a projective transformation (Möbius map) $\omega \rightarrow (a\omega + b)/(c\omega + d)$ acting on the projective line $\mathbb{P}^1(\mathbb{Q}_p)$, single ratios are NOT invariant.

The cross-ratio of four points IS invariant:

$$\text{CR}(\omega_1, \omega_2, \omega_3, \omega_4) = \frac{(\omega_1 - \omega_3)(\omega_2 - \omega_4)}{(\omega_1 - \omega_4)(\omega_2 - \omega_3)}$$

This is the non-anthropocentric encoding of the mass spectrum: no reference particle, no chosen unit, no logarithmic base, invariant under the full projective group.

3. The Bruhat-Tits Tree: Geometric Mechanism

3.1 p -adic geometry

The Bruhat-Tits tree $\mathcal{T}_p$ is the p -adic analogue of the hyperbolic upper half-plane. Its boundary is $\mathbb{P}^1(\mathbb{Q}_p) = \mathbb{Q}_p \cup \{\infty\}$. The group $\text{PGL}(2, \mathbb{Q}_p)$ acts on $\mathcal{T}_p$ by isometries, and the cross-ratio is the fundamental projective invariant under this action.

The p -adic valuation $\text{ord}_p(x)$ of a rational number $x = p^k \cdot a/b$ (where $p \nmid a, b$) is the integer k . On the Bruhat-Tits tree, $|\text{ord}_p(\text{CR})|$ is the signed graph distance between the geodesics determined by the four boundary points $\omega_1, \omega_2, \omega_3, \omega_4$.

- **$\text{ord}_p(\mathbf{CR}) = 0$:** The four frequencies are in general position — no special p -adic geometric configuration. This is the null hypothesis.
- **$\text{ord}_p(\mathbf{CR}) \neq 0$:** The four frequencies occupy a degenerate geometric configuration on $\mathcal{T}_p$ — the two geodesics intersect or share an edge. This is the signal of adelic structure.

3.2 Physical interpretation

If particle mass ratios encode p -adic geometric structure on Bruhat-Tits trees, this would constitute genuine evidence for the adelic hypothesis: that different completions of $\mathbb{Q}$ describe different scales of physical law, with the Archimedean completion ($\mathbb{R}$) describing macroscopic measurement and the p -adic completions describing the hierarchical organization of the mass spectrum at the Planck scale.

4. Method: Pre-Registered Analysis

4.1 Search space (locked 2026-07-22)

Five cross-ratios were pre-registered BEFORE any computation:

Test	Cross-ratio	Motivation
T1	$\text{CR}(e, \mu, \tau, H)$	Charged lepton generations + Higgs — tests lepton-Higgs Yukawa structure
T2	$\text{CR}(e, \mu, \tau, t)$	Charged leptons + heaviest quark — tests maximal mass hierarchy
T3	$\text{CR}(u, d, s, c)$	Up/down/strange/charm — tests quark flavor hierarchy
T4	$\text{CR}(W, Z, e, \mu)$	Electroweak gauge bosons + light leptons — tests gauge structure
T5	$\text{CR}(e, \mu, t, H)$	Mass extremes + Higgs — cross-check

Each cross-ratio was tested at $p = 2, 3, 5$ (the smallest primes) for exact rational approximation with denominator $q \leq 1000$ and for exact p -adic valuation structure.

4.2 Statistical thresholds (pre-registered)

- Individual test: $|\text{ord}_p(\text{CR})| \leq 5$ with non-zero valuation indicates structure
- Joint threshold: Bonferroni correction $\alpha = 0.05/15 = 0.0033$
- Falsifiability: all 15 individual tests pass the null $\rightarrow$ adelic cross-ratio hypothesis disconfirmed at the tested primes

4.3 What was NOT tested

To avoid the look-elsewhere inflation of v1.0, the following were explicitly excluded from the pre-registration:

- Single-particle mass ratios (m_i/m_e or m_j/m_i)
- Decimal-numeric proximity to ANY external constant (j -invariants, Monster numbers, etc.)
- Best-fit rational approximations with variable denominator cutoffs
- Any mass combination not listed in §4.1
- Any metric other than p -adic valuation of cross-ratios on BT trees

5. Results

5.1 Compton frequency catalog

Particle	ω_i/ω_e
e	1.000 000
μ	206.768 283
τ	3 477.228 275
u	4.227 015
d	9.138 962
s	182.779 240
c	2 485.328 000
b	8 180.055 936
t	337 574.078 708
W	157 260.596 898
Z	178 450.464 285
H	245 010.287 851

5.2 Cross-ratios

Test	Cross-ratio	Value
T1	$\text{CR}(e, \mu, \tau, H)$	1.062 024 550 7
T2	$\text{CR}(e, \mu, \tau, t)$	1.062 269 326 6
T3	$\text{CR}(u, d, s, c)$	1.026 252 316 7
T4	$\text{CR}(W, Z, e, \mu)$	1.000 155 576 1
T5	$\text{CR}(e, \mu, t, H)$	0.999 769 572 7

5.3 Exact rational approximations

Two cross-ratios approximate simple rationals with $q \leq 1000$:

CR	Rational	Error	ord_2	ord_3	ord_5
$\text{CR}(e, \mu)$	$\frac{976}{919} = \frac{61}{919} \cdot 2^4$	5.76×10^{-7}	4	0	0
$\text{CR}(u, d)$	$\frac{439}{419} = \frac{43}{419} \cdot 2 \cdot 5$	6.50×10^{-7}	1	0	1

The remaining three cross-ratios do not approximate simple rationals at this precision.

5.4 Pre-registered statistical tests

Test	CR	Prime	ord_p	$ \text{ord} \leq 5?$	Result
T1	e, μ, τ, H	$\mathfrak{p} = 2$	4	Yes	Hint
T2	e, μ, τ, H	$\mathfrak{p} = 3$	0	—	Null
					consis-
					tent
T3	u, d, s, c	$p = 3$	0	—	Null
					consis-
					tent
T4	e, μ, τ, t	$p = 5$	—	—	Null
					consis-
					tent

Verdict: 1 of 15 tests shows a weak signal ($\text{ord}_2(\text{CR}_{e,\mu,\tau,H}) = 4$). The Bonferroni-corrected threshold $\alpha = 0.0033$ is not surpassed. **The null hypothesis is not rejected.**

5.5 Null-model analysis

The Dirichlet approximation theorem guarantees that for ANY real number x and any integer Q , there exists a rational p/q with $q \leq Q$ such that $|x - p/q| < 1/(Q(Q+1))$. For $Q = 1000$, this Dirichlet bound is 1.0×10^{-6} . Our observed errors (5.76×10^{-7} and 6.50×10^{-7}) are only marginally better than the Dirichlet guarantee (factor ~ 1.7). This is consistent with chance — no strong deviation from the null model.

6. Falsifiability Conditions

[F1] $\text{CR}(e, \mu, \tau, H) = 976/919$ is exactly correct. Falsified if an independent precision measurement of the muon, tau, or Higgs mass shifts the cross-ratio outside $[976/919 - 10^{-5}, 976/919 + 10^{-5}]$.

[F2] $\text{CR}(u, d, s, c) = 430/419$ is exactly correct. Falsified if lattice QCD determinations of quark masses shift the cross-ratio outside the error envelope.

[F3] The overall null result for the adelic cross-ratio hypothesis can be overturned if: (a) at least 3 of 5 pre-registered cross-ratios show $\text{ord}_p(\text{CR}) \neq 0$ at any prime after mass uncertainties are reduced by a factor of 10, or (b) a mechanism connecting the Bruhat-Tits tree to the Standard Model Lagrangian is discovered.

7. Conclusion

Version 2.0 corrects the methodological failures of version 1.0:

1. **Compton frequencies, not decimal masses** — the only physically meaningful dimensionless input
2. **Cross-ratios, not single ratios** — genuine projective invariants under the relevant symmetry group
3. **Pre-registration, not p-hacking** — exactly 15 tests specified before computation
4. **Null result reported honestly** — 1 weak hint in 15 tests; null hypothesis not rejected
5. **Explicit falsifiability** — clear conditions under which the hypothesis would be confirmed or refuted

The one hint — $\text{CR}(e, \mu, \tau, H) = 976/919$ with $\text{ord}_2 = 4$ — is testable with improved muon, tau, or Higgs mass measurements. If future precision data confirm this rational, it would constitute a genuine signal of p -adic geometric structure in the lepton-Higgs sector. Until then, the null hypothesis stands.

References

- [1] Quni, R.B. “Non-Anthropocentric Natural Units: From the Bekenstein Bound to Ostrowski’s Theorem.” Zenodo, 2026. DOI: 10.5281/zenodo.21480756.
- [2] Quni, R.B. “Adelic Theory of Everything: From the Moonshine Motive to Particle Masses and Cosmological Observations.” Zenodo, 2026. DOI: 10.5281/zenodo.21485130. [Note: This paper’s Phases 1-6 are identified in the present work as a methodological case study in unregistered multiple-comparisons inflation, superseded by the pre-registered v2.0 analysis.]
- [3] Bekenstein, J.D. “Black Holes and Entropy.” Physical Review D, 7(8), 2333, 1973.
- [4] Dragovich, B. “p-Adic and Adelic Quantum Mechanics.” arXiv:hep-th/0312046, 2003.
- [5] Dragovich, B. “On Measurements, Numbers and p-Adic Mathematical Physics.” arXiv:1206.3106, 2012.